

Vansh

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Summary

Full-Stack and Machine Learning Engineer specializing in scalable MERN stack web applications and high-accuracy AI models. Experienced in building real-time AI solutions, optimizing data pipelines, and architecting robust cloud infrastructure to drive measurable performance improvements.

Education

B.Tech in Computer Science, VIT Bhopal *Sept 2023 – Aug 2027 | 8.79 CGPA*
Class XII, Floradale School, Ganganagar, Rajasthan *84.8%*
Class X, D.A.V. Public School, Suratgarh, Rajasthan *95.33%*

Skills

Languages: Java, Python, C++, JavaScript, TypeScript

Web Technologies & Libraries: React, Node.js, Express.js, Next.js, Flask, FastAPI, HTML/CSS, Tailwind CSS, MUI, Shadcn, Socket.io, JWT

Machine Learning & AI: TensorFlow, PyTorch, Pandas, NumPy, Scikit-learn, Large Language Models (LLMs), Prompt Engineering

Databases: MongoDB, MySQL, PostgreSQL, Oracle Database, Supabase, Prisma

Cloud & DevOps: AWS, Oracle Cloud Infrastructure (OCI), Docker, CI/CD

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), REST APIs

Experience

Web Development Intern — KL LAB *May 2025 – Jul 2025 | Remote*

- Implemented secure and scalable REST APIs using Node.js and Express.js, implementing JWT-based authentication that protected 20+ backend endpoints and reduced unauthorized access attempts by 30%.
- Architected optimized MongoDB schemas via Mongoose, accelerating data retrieval efficiency for user, blog, and comment workflows by 40% and reducing query latency to under 50ms
- Deployed backend systems to cloud infrastructure, collaborating with a 4-person frontend team to achieve seamless full-stack integration and maintain 99.9% uptime post-launch.

Technical Team Core Member — AWS Club, VIT Bhopal *Oct 2024 – Jul 2025 | Bhopal, MP*

- Spearheaded 5+ technical workshops on AWS fundamentals for 100+ members, driving a 20% increase in club engagement and leading team-based cloud architecture projects.

Projects

AI-Powered ATS Resume Screener | **GitHub** | **Live Demo** *Next.js 16, React, Shadcn, Supabase, Prisma, PostgreSQL, TypeScript, LLMs*

- Engineered an enterprise ATS simulation platform that evaluates resumes concurrently against Legacy (Taleo, Workday) and Modern (Greenhouse, Eightfold) systems, utilizing distinct parser strictness and semantic evaluation rules.
- Halved API latency and token consumption by consolidating the dual-persona evaluation into a single structured LLM prompt, achieving sub-5-second processing times for complex multi-system analysis.
- Created a high-availability fallback engine cascading from primary Gemini models to Groq (Llama 3.3 70B → Llama 3.1 8B) with exponential back-off, guaranteeing near 100% platform uptime during provider rate limits.
- Designed a highly concurrent data architecture featuring a fail-open Prisma 7 rate-limiting system and unified PostgreSQL logging transactions, deployed behind edge-protected Clerk v6 middleware.

Real-Time Network Intrusion Detection System | **GitHub** | **Live Demo** *Python, XGBoost, React, WebSockets, Docker*

- Developed a high-performance XGBoost Binary Classifier trained on 421,959 cleansed network flows from the CIC-DDoS2019 dataset, achieving 99.85% accuracy in distinguishing malicious from benign traffic.
- Streamlined a custom Python data pipeline using Scapy to extract 15 distinct statistical features from live network flows for real-time edge inference.
- Established a zero-latency SOC (Security Operations Center) dashboard using React and WebSockets, pushing threat alerts and live packet graphs to the client in under 2 seconds.
- Orchestrated a container microservices environment utilizing Docker Compose, isolating the Node.js, React, and Python engines to reduce local deployment time by 80% (under 2 minutes).

SureBet | **GitHub** | **Live Demo** *React, Node.js, Express.js, MongoDB, Socket.io*

- Produced a real-time sports arbitrage discovery platform scanning 15+ sportsbooks, featuring instant data updates via WebSockets to deliver high-value betting opportunities with sub-200ms latency.
- Programmed advanced filtering and sorting algorithms across 10+ data points, enabling users to isolate high-yield opportunities 50% faster.
- Pioneered a full-stack architecture using Node.js, Express, and MongoDB, facilitating scalable backend logic and providing secure data handling for over 100+ data points and improved data retrieval speeds 2x.
- Planned a virtualized data table within the dashboard with sortable columns and live stake breakdowns, utilizing virtual scrolling to render 5,000+ data rows without frame drops, improving client-side load speeds by 60%.

Certifications

- Oracle Cloud Infrastructure 2025 Certified DevOps Professional
- Oracle Cloud Database Services 2025 Certified Professional
- MATLAB

- AWS Cloud Club - Machine Learning Camper
- The Bits and Bytes of Computer Networking - Google